

Backend Engineer Intern at Pharos Indonesia: Profiling Feature and Whatsapp Blast Integration

During my internship at Pharos Indonesia, I developed backend features to streamline sales promotions and customer profiling. I designed the database schema and implemented Django-based web forms to manage doctor and cashback receiver data. I also built a verification feature where sales teams could upload and review promotional screenshots, ensuring accountability. Additionally, I integrated the WhatsApp Blast system by placing broadcast options (single or multiple items) and message templates directly into the platform. These contributions improved data reliability and made promotional campaigns more efficient for the sales team.

What I learned: This project taught me how to build end-to-end backend features, balancing database design, web logic, and frontend integration within a real company workflow.

Year Accomplished
2024

Role/Position
Backend Engineer

Publication Link
[Link](#)

Type
Internship
Project/Engineer

The screenshot displays the 'Data Penerima -klinik' form, which is used for entering patient data. The form includes fields for patient name, address, phone number, and bank information. Below the form are buttons for 'Edit Data Penerima' and 'Delete Penerima Klinik'. To the right of the form is a list of products, including 'ROVERTON 15MG/5ML SYR 60ML', 'LANOPRAZOLE 30 MG 50'S (PH)', and 'DELFRIN ORANGES VANILLA 60ML'.

The screenshot shows the 'Pilih Penerima Blast' interface, which is used for selecting recipients for a WhatsApp blast. It features a search bar at the top and a list of products below. The products include 'IPI VITAMIN B COMPLEX 90 TAB', 'AMOXICILLIN 500MG 100'S [GEN]', 'LAXANA 5MG 10STRIP@10'S', 'DEXAMETHASONE 0.5MG TAB 100'S', 'ZANTIFAR 150MG TAB 50'S', 'ACYCLOVIR 200MG 100'S [GEN]', and 'CETIRIZINE HCL 10MG 30'S OGB'.

Rainfall Estimation Based on Convolutional Neural Network – Long Short Term Memory (CNN-LSTM) with Surveillance Camera Video Input

For my bachelor thesis, I researched how temporal modeling with CNN-LSTM networks can improve rainfall estimation using surveillance camera video input. I aligned rainfall ground-truth data from rain gauges with synchronized CCTV footage, then built a dataset with six intensity levels (class 0 = no rain to class 5 = heavy rain). To ensure balanced training, I applied equal class distribution and controlled time-of-day ratios (morning, afternoon, evening, night = 4:1:4:1). I compared grayscale and RGB inputs, finding grayscale allowed larger dataset usage due to lighter data size. I also demonstrated that longer input durations (3 seconds at 5 fps) yielded better results than shorter, high-FPS sequences.

What I learned: This project strengthened my skills in dataset balancing, temporal modeling, and preprocessing, teaching me how to design robust AI systems under real-world conditions.

Year Accomplished
2025

Role/Position
AI Researcher

Publication Link
[Link](#)

Type
Bachelor Thesis Project



SAPA: Sarana Pembelajaran Bahasa Isyarat [Bangkit Project]

As part of the Bangkit Academy capstone project, my team and I created SAPA, a mobile application to support learning of SIBI (Sistem Isyarat Bahasa Indonesia). I contributed as the AI Engineer by designing and implementing the hand sign recognition system. I developed a preprocessing pipeline including image resizing, augmentation, bounding box extraction, and hand skeleton detection. I then trained a neural network-based model, achieving ~95% accuracy in hand sign classification. To make the system production-ready, I converted the model into TensorFlow Lite for seamless Android integration. Working with cloud, mobile, and ML teammates, I ensured the AI system functioned smoothly within the app.

What I learned: I gained hands-on experience in deploying machine learning models to mobile platforms and collaborating in a cross-functional team to deliver a user-focused application.

Year Accomplished
2024

Role/Position
AI Engineer

Publication Link
[Link](#)

Type
Bangkit Capstone
Project/Group

